

IS6200/IS6143 – Lab 2 Document

General guidelines

Naming

- When creating the filenames for your programs, please use the convention specified by your instructor. Otherwise, for programs that consist of just one file, name the file *first_last_program.py* where *first_last* specifies your first and last name and *program* specifies the name of the program. For programs that have multiple files, store the files in a folder named *first_last_program*.
- When creating names for variables and functions, please use the guidelines and recommendations specified by your instructor.

User interfaces

- You should think of the user interfaces that are shown for the projects as starting points. If you can improve on them, especially to make them more user-friendly, by all means do so.

Specifications

- You should think of the specifications that are given for the projects as starting points. If you have the time to enhance the programs by improving on the starting specifications, by all means do so.

Top-down development

- Always start by developing a working version of the program for a project. That way, you'll have something to show for your efforts if you run out of time. Then, you can build out that starting version of the program until it satisfies all of the specifications.
- You should use top-down coding and testing as you develop your programs. You might also want to sketch out a hierarchy chart for each program as a guide to your top-down coding.

Files supplied by your instructor

- Some of the projects require starting text, CSV, or binary files. These files are identified in the specifications for the projects, and your instructor should make these starting files available to you.

Project 2-1: Letter Grade Converter

Create a program that converts number grades to letter grades.

Console

```
Letter Grade Converter

Enter numerical grade: 90
Letter grade: A

Continue? (y/n): y

Enter numerical grade: 88
Letter grade: A

Continue? (y/n): y

Enter numerical grade: 80
Letter grade: B

Continue? (y/n): y

Enter numerical grade: 67
Letter grade: C

Continue? (y/n): y

Enter numerical grade: 59
Letter grade: F

Continue? (y/n): n

Bye!
```

Specifications

- The grading criteria is as follows:

A	88-100
B	80-87
C	67-79
D	60-66
F	<60
- Assume that the user will enter valid integers for the grades.
- The program should continue only if the user enters “y” or “Y” to continue.

Project 2-2: Tip Calculator

Create a program that calculates three options for an appropriate tip to leave after a meal at a restaurant.

Console

```
Tip Calculator

Cost of meal: 52.31

15%
Tip amount:    7.85
Total amount:  60.16

20%
Tip amount:    10.46
Total amount:  62.77

25%
Tip amount:    13.08
Total amount:  65.39
```

Specifications

- The program should calculate and display the cost of tipping at 15%, 20%, or 25%.
- Assume the user will enter valid data.
- The program should round results to a maximum of two decimal places.

Project 2-3: Change Calculator

Create a program that calculates the coins needed to make change for the specified number of cents.

Console

```
Change Calculator

Enter number of cents (0-99): 99

Quarters: 3
Dimes:    2
Nickels:  0
Pennies:  4

Continue? (y/n): y

Enter number of cents (0-99): 55

Quarters: 2
Dimes:    0
Nickels:  1
Pennies:  0

Continue? (y/n): n

Bye!
```

Specifications

- The program should display the minimum number of quarters, dimes, nickels, and pennies that one needs to make up the specified number of cents.
- Assume that the user will enter a valid integer for the number of cents.
- The program should continue only if the user enters “y” or “Y” to continue.

Project 2-4: Shipping Calculator

Create a program that calculates the total cost of an order including shipping.

Console

```
=====
Shipping Calculator
=====
Cost of items ordered: 49.99
Shipping cost:         7.95
Total cost:            57.94

Continue? (y/n): y
=====
Cost of items ordered: -65.50
You must enter a positive number. Please try again.
Cost of items ordered: 65.50
Shipping cost:         9.95
Total cost:            75.45

Continue? (y/n): n
=====
Bye!
```

Specifications

- Use the following table to calculate shipping cost:

COST OF ITEMS	SHIPPING COST
=====	
< 30.00	5.95
30.00-49.99	7.95
50.00-74.99	9.95
> 75.00	FREE

- If the user enters a number that's less than zero, display an error message and give the user a chance to enter the number again.

Project 2-5: Table of Powers

Create a program that displays a table of squares and cubes for the specified range of numbers.

Console

```
Table of Powers

Start number: 90
Stop number: 100

Number      Squared      Cubed
=====
90 8100      729000
91 8281      753571
92 8464      778688
93 8649      804357
94 8836      830584
95 9025      857375
96 9216      884736
97 9409      912673
98 9604      941192
99 9801      970299
100 10000     1000000
```

Specifications

- The formulas for calculating squares and cubes are:

```
square = x ** 2
cube = x ** 3
```

- Use tabs to align the columns.
- Assume that the user will enter valid integers.
- Make sure the user enters a start integer that's less than the stop integer. If the user enters a start integer that's greater than the stop integer, display an error message and give the user a chance to enter the integers again.